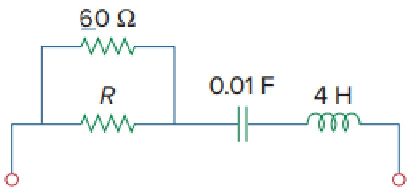
Chapter 8, Problem 13P

Problem

For the circuit in Fig. 8.68, calculate the value of *R* needed to have a critically damped response.

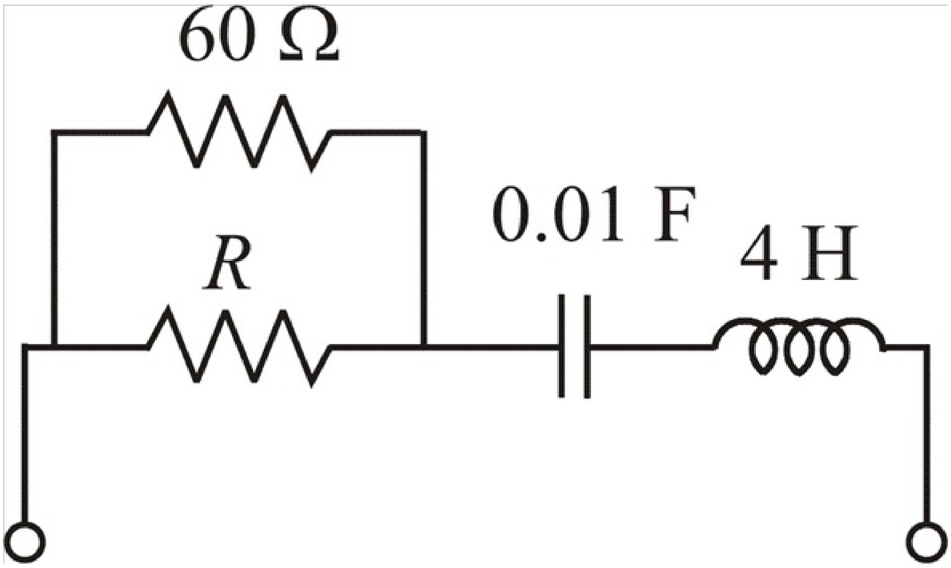
Figure 8.68:



Step-by-step solution

Step 1 of 3

The given circuit is,



Step 2 of 3

Let, $R_1 = (60 \parallel R)$

$$R_1 = \frac{60R}{60+R} \text{.................. (1)}$$

At critically damped,

$$\omega_0 = \frac{R_1}{2L} = \frac{1}{\sqrt{LC}}$$

$$R_1 = \frac{2L}{\sqrt{LC}}$$

$$R_1 = \frac{2 \times 4}{\sqrt{4 \times 0.01}}$$

$$R_1 = 40\Omega \text{.................... (2)}$$

Step 3 of 3

From equations (1) and (2), we can write that

$$\frac{60R}{60+R}=40$$

$$60R=2400 + 40R$$

$$R=\frac{2400}{20}$$

$$\boxed{R=120\Omega}$$